1. task 1

$(\lambda x.\lambda y.y)(\lambda a.aa)$

$\rightarrow (\lambda y.y)$

$((\lambda x.\lambda y.xy)(\lambda z.\lambda w.w))(\lambda a.aa)$

$\rightarrow (\lambda y.(\lambda z.\lambda w.w)y)(\lambda a.aa)$

$\rightarrow (\lambda z.\lambda w.w)(\lambda a.aa)$

$\rightarrow \lambda w.w$

2. task 2

sum $:= \lambda t.$ fold $(+)(\lambda x.x)t$

sum_leaves $:= \lambda t.$ fold $(\lambda a.\lambda b.\lambda c.(+)bc)(\lambda x.x)t$

map $:= \lambda m.$ fold $(\lambda v.\lambda l.\lambda r.$ node $(mv)lr)(\lambda v.$ leaf $(mv))$

size $:=$ fold $(\lambda v.\lambda l.\lambda r + (+c_1 l)r)(\lambda v.c_1)$

3. task 3

3.1. inference rules:

$$\frac{t_1 \to t_{1'}}{t_1 t_2 \to t_{1'} t_2}$$

$(\lambda x.t_{12})t_2 \longrightarrow [x \mapsto t_2]t_{12}$

3.2. evaluate:

3.2.1. cbv

0. $(\lambda f.\lambda v.vv)((\lambda x.x)(\lambda y.\lambda z.y))((\lambda a.aa)(\lambda y.y))$
1. $(\lambda f.\lambda v.vv)(\lambda y.\lambda z.y)((\lambda a.aa)(\lambda y.y))$
2. $(\lambda v.vv)((\lambda a.aa)(\lambda y.y))$
3. $(\lambda v.vv)((\lambda y.y)(\lambda y.y))$
4. $(\lambda v.vv)(\lambda y.y)$
5. $(\lambda y.y)(\lambda y.y)$
6. $(\lambda y.y)$

3.2.2. cbn

0. $(\lambda f.\lambda v.vv)((\lambda x.x)(\lambda y.\lambda z.y))((\lambda a.aa)(\lambda y.y))$
1. $(\lambda v.vv)((\lambda a.aa)(\lambda y.y))$
2. $((\lambda a.aa)(\lambda y.y))((\lambda a.aa)(\lambda y.y))$
3. $((\lambda y.y)(\lambda y.y))((\lambda a.aa)(\lambda y.y))$
4. $(\lambda y.y)((\lambda a.aa)(\lambda y.y))$
5. $(\lambda a.aa)(\lambda y.y)$
6. $(\lambda y.y)(\lambda y.y)$
7. $(\lambda y.y)$

3.3. comparison

upsides cbn:

- unused arguments dont have to be evaluated
- if an argument diverges but is not used it can be ignored and evaluation can continue

upsides cbv:

- every argument only has to be evaluated once, even if it is used more than once inside a function